

Mehak Wadhwa

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PROFESSIONAL SUMMARY

Final-year B.Tech CSE (Bioinformatics) student with hands-on experience in software development, REST API integration, full-stack applications, and data-driven systems. Experienced with Python, Java, React, TypeScript, FastAPI, SQL, Git, and Docker. Built backend services, interactive bioinformatics applications, and data analysis pipelines. Seeking entry-level Software Development Engineer opportunities.

EDUCATION

Vellore Institute of Technology (VIT)

B.Tech in Computer Science and Engineering (Bioinformatics)

July 2023 – 2027

Vellore, Tamil Nadu

- **CGPA:** 8.64
- **Leadership:** Secretary of Geospatial club(2025-2026); Event Coordinator for Institute Innovation Council(2024)

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, C/C++, HTML/CSS

AI & Machine Learning: TensorFlow, Scikit-Learn, Neural Networks, Deep Learning, Predictive Modeling, Data Preprocessing

Web & APIs: REST APIs, Frontend Development, Backend Integration, API Debugging, Payload Handling

Developer Tools: Git, GitHub, Linux (WSL), Version Control, Data Engineering Pipelines

Bioinformatics: Pandas, NetworkX, Biopython, Matplotlib, Data Analysis

Languages: English (Fluent), Hindi (Native), Spanish(Beginner)

EXPERIENCE

Infinite Locus

SDE Intern

May 2026 – July 2026

Gurugram / India

- Resolved frontend defects across the Master, Instructor, and Student web portals of Ultimate Drivers Canada, improving application usability and reliability.
- Integrated and debugged REST APIs by inspecting request/response payloads, status codes, and frontend-backend data flow.
- Collaborated in an Agile development environment using Git feature branches, commits, merges, and version control workflows.

PROJECTS

SARS-CoV-2 Genomic Drift Analysis | *Python, Biopython, Matplotlib*

- Engineered a bioinformatics pipeline to analyze evolutionary changes by comparing 2020 reference sequences with 2026 variants.
- Parsed FASTA genomic sequences and computed GC-content stability metrics, identifying a measurable 1.5% shift in molecular stability.

Gene-Disease Network Explorer | *Python, Biopython, NetworkX*

- Developed a bioinformatics application to explore gene-disease associations using biological databases and genomic data.
- Built gene-disease interaction networks to visualize relationships and identify disease-associated genes such as BRCA1.

National Data Analytics Pipeline | *Python, SQL, Data Engineering* | [GitHub](#)

March 2026

- Architected an automated processing pipeline for cleaning and analyzing UDISE and UIDAI datasets using Python and SQL.
- Generated state-wise visualizations of enrollment distributions for 2022–23 to identify regional educational trends across Indian states.

PUBLICATIONS & CERTIFICATIONS

Publication

- **Intelligent Hydraulic System Monitoring via Masked Attention Pooling and Threshold-Specific Ordinal Regression for Real-Time Fault Detection and Cloud-Hosted Anomaly Detection on Real-Time Streaming Data**
Submitted to Turkish Journal of Electrical Engineering & Computer Sciences (Manuscript No. TJEECS-2025-01804).

Training

- Artificial Intelligence – SmartBridge in collaboration with Google for Developers